Wei-Chen Liu

Automotive AI Perception & Multi-Sensor System Architect
Taipei, Taiwan

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R&D 100 Award Winner | SOTIF (ISO 21448) Professional

Professional Summary

Automotive AI engineer with 8+ years of experience spanning multi-sensor perception, dataset architecture, and real-world autonomous deployment.

Specialized in camera, thermal imaging, LiDAR, and 4D radar perception systems for ADAS and intelligent mobility. Experienced across the full technical pipeline including sensor calibration, dataset construction, perception algorithm development, system integration, and field validation.

Led development of Taiwan's representative multi-sensor road scene dataset **FORMOSA**, and the **VisionSafe smart cockpit system**, which received the **2025 R&D 100 Award**.

Technical Stack

Programming

C++, Python

AI / Machine Learning

PyTorch, ONNX, TensorRT

Computer Vision

OpenCV, Deep Learning Perception

Robotics & Middleware

ROS, RViz

Sensors

Camera, Thermal Imaging, LiDAR, 4D Radar, GPS/IMU

Hardware Platforms

NVIDIA Jetson, Edge AI SoC

Vehicle Systems

CAN Bus integration, MDVR systems

Core Expertise

Automotive AI Perception

- Multi-sensor perception system design
- AI-ADAS algorithm development
- Sensor fusion and perception pipeline optimization
- Real-world performance validation

Multi-Sensor Dataset Engineering

- Dataset architecture design
- Multi-sensor calibration and synchronization
- Cross-modal object-level annotation frameworks
- Large-scale deep learning dataset construction

System Integration

- Perception-to-control pipeline integration
- Autonomous vehicle validation
- Edge AI deployment
- AR-HUD safety visualization systems

Professional Experience

Institute for Information Industry (III), Taiwan

Section Manager – Intelligent Mobility & AI Systems

Mar 2024 – Present

- Lead development of automotive AI perception and smart cockpit systems
- Manage cross-functional teams covering perception algorithms, data engineering, and system integration
- Oversee multi-sensor perception projects integrating camera, thermal imaging, LiDAR, and 4D radar
- Drive collaboration with industry partners for commercial vehicle AI-ADAS deployment

Software Engineer – Automotive AI & Perception Systems

Oct 2017 – Feb 2024

- Developed perception algorithms for automotive AI systems
- Designed and implemented multi-sensor datasets for autonomous driving
- Conducted real-world validation across diverse traffic and environmental conditions
- Participated in autonomous vehicle deployment and intelligent mobility demonstrations

Selected Projects

FORMOSA Dataset

Taiwan Multi-Sensor Autonomous Driving Dataset

Role: Primary Designer & Core Executor

Designed and implemented Taiwan’s representative multi-sensor dataset supporting perception research and industry collaboration.

Key contributions:

- Multi-sensor dataset architecture design
- Sensor calibration and synchronization
- Cross-sensor object-level annotation framework

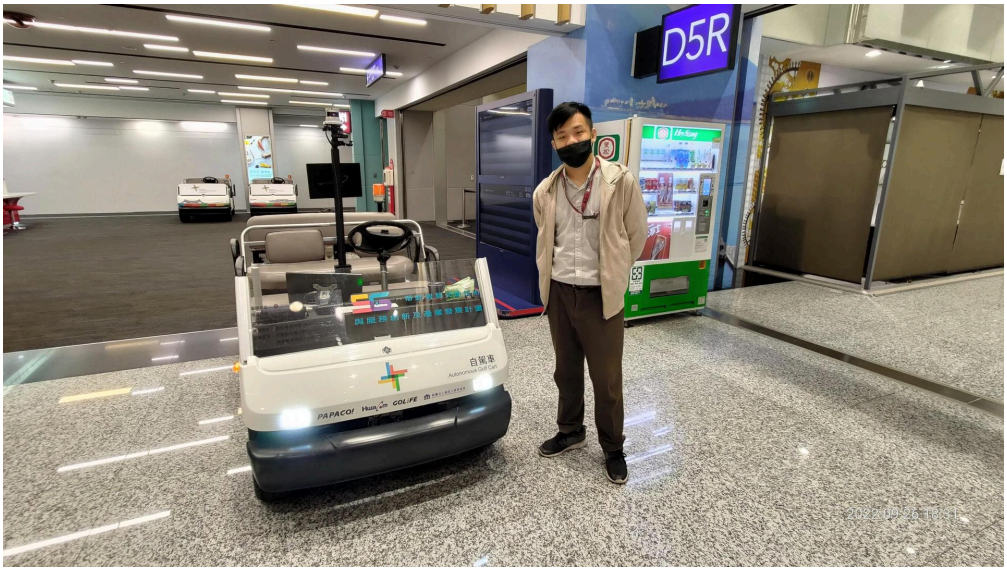
Sensor modalities:

- Camera
- Thermal Imaging
- LiDAR
- 4D Radar
- GPS/IMU

Applications:

- Autonomous driving perception research
- Multi-sensor AI model training
- Industry collaboration projects

Taoyuan International Airport Autonomous Shuttle



Role: Technical Lead

Real-world autonomous shuttle validation deployed at Taoyuan International Airport.

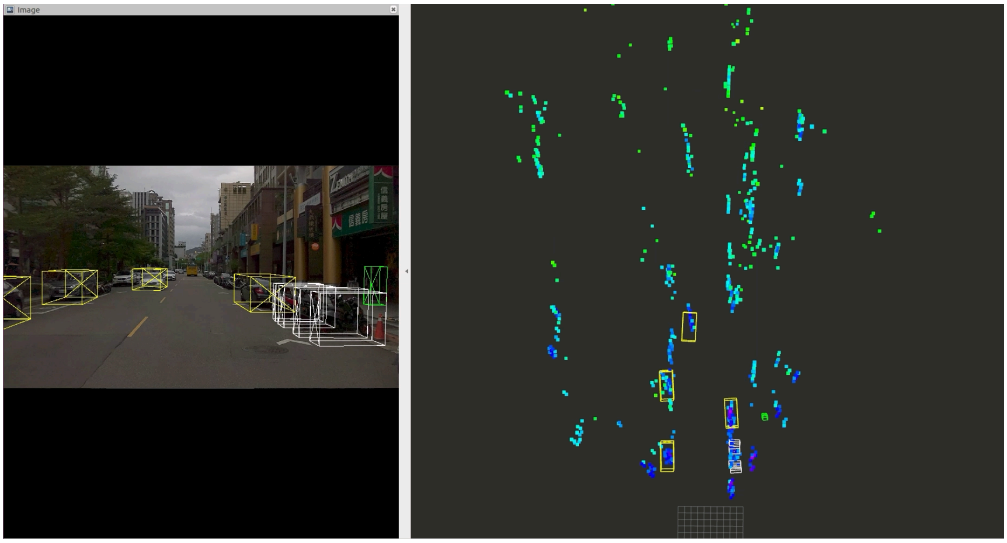
Key metrics:

- Localization latency < 40 ms
- Perception-to-control processing < 60 ms
- 3,000+ passenger rides completed during a 3-month pilot

Technologies integrated:

- AI perception systems
- High-definition mapping
- Autonomous navigation and control algorithms

4D Radar AI Perception Collaboration



Collaborated with industry partners to develop AI perception technology for 4D radar.

Key work:

- Radar dataset construction
- Object detection model development
- Evaluation framework for radar perception

Applications:

- Commercial vehicle ADAS systems

VisionSafe Smart Cockpit System

R&D 100 Award Winner (2025)

AI-powered smart cockpit solution designed for commercial vehicles.

System features:

- AI-ADAS perception modules
- AR-HUD real-time safety visualization
- Multi-sensor perception combining thermal and visible cameras

Impact:

- Improves safety in blind-spot and adverse weather conditions
- Demonstrated at **CES 2025**
- Recognized with **R&D 100 Award**

Publications & Honors

R&D 100 Award (2025)
VisionSafe Smart Cockpit Solution

SOTIF (ISO 21448) Professional Certification
SGS TÜV Saar

SGS TÜV SAAR

Wei-Chen Liu

is authorized to bear the title of

SOTIF PRO by SGS TÜV

SOTIF Professional

The following qualifications are confirmed through the title:

Extensive knowledge of contents of the ISO DIS 21448 automotive standard

Extensive knowledge regarding the interpretation of the requirements resulting from ISO DIS 21448

Extensive knowledge of the methodology to implement the requirements according to ISO DIS 21448

The certification is based on the following qualification measures:

Attendance of the specific technical seminars of SGS-TÜV Saar to interpret and apply ISO DIS 21448 (modules S1+S2) or comparable training activities

Successful SOTIF PRO exam

Certificate issued on: ***November 25, 2022***
This certificate is valid for 3 years

Marcus Rau
Global Operations & Training Manager

Martin Schmidt
Global Manager

SGS-TÜV Saar GmbH, Certification Body Functional Safety & Cyber Security, Hofmannstr. 50, 81379 Muenchen, Germany

ERTIFICATE

IEEE ICIEB 2021

All-Weather Automotive Perception:
AI-ADAS using Fusion of Thermal Imaging and RADAR Sensors

Education

M.S., Electrical and Electronics Engineering
National Taiwan Normal University

B.S., Electrical and Electronics Engineering
National Taiwan Normal University